Chapter 13 Homework

Part A Multiple choice (Select the one that is best in each case. 1 point/question)

1. Which of the following substances is least likely to dissolve in water?
A) $CH_3(CH_2)_9CHO$
B) $HOCH_2CH_2OH$
C) $CH_3(CH_2)_9CH_2OH$
D) $CHCl_3$
E) CCl_4

2. How does the solubility of a gas in a solvent depend on pressure and temperature?
A) Increasing the partial pressure of the gas while increasing the temperature increases the solubility of the gas.
B) Decreasing the partial pressure of the gas while decreasing the temperature increases the solubility of the gas.
C) Increasing the partial pressure of the gas while decreasing the temperature increases the solubility of the gas.
D) Decreasing the partial pressure of the gas while increasing the temperature increases the solubility of the gas.
E) Gas solubility is unaffected by pressure or temperature.

3. Which of the following favors the solubility of an ionic solid in a liquid solvent?
A) a large magnitude of the solvation energy of the ions
B) a small magnitude of the lattice energy of the solute
C) a large polarity of the solvent
D) all of the above
E) none of the above

4. At a particular temperature the solubility of O_2 in water is 0.590 g/L at an oxygen pressure of around 15.2 atm. What is the Henry's law constant for O_2 (in units of mol/L-atm)
A) 3.88×10^{-2}
B) 8.24×10^{2}
C) 2.80×10^{-1}
D) 1.21×10^{-3}
E) None is correct.

5. As the number of solute particles in a given volume of solution increases,
A) the boiling point will increase and the vapor pressure will increase.
B) the freezing point will decrease and the vapor pressure will decrease.
C) the freezing point will increase and the vapor pressure will increase.
D) the boiling point will decrease and the vapor pressure will decrease.
E) the osmotic pressure will decrease and the lattice energy will increase.

6. In a 0.1 *M* solution of NaCl in water, which one of the following will be closest to 0.1?
A) The molality of NaCl.
B) The mass fraction of NaCl.
C) The mass percentage of NaCl.
D) The mole fraction of NaCl.
E) All of these are about 0.1.

7. What is the mass percentage of Na_2CO_3 in a 1.92 *m* aqueous solution?
A) 0.160% B) 16.9% C) 99.5% D) 20.4% E) 19.2%

8. What is the mole fraction of urea, CH_4N_2O, in an aqueous solution that is 37% urea by mass?
A) 0.85 B) 0.37 C) 0.54 D) 0.66 E) 0.15

9. Which of the following concerning the topic of concentration is/are correct?
I. A 5.0 *m* HCl solution consists of 5.0 moles HCl dissolved in 1.0 kg of solution.
II. When calculating a mass percentage, the "mass of solution" is equal to the sum of the masses of all solution components.
III. The mole fraction of a component in a solution may be greater than 1.
A) I only B) II only C) III only D) II and III E) None of the above

10. 10.0 g of an alkali metal chloride salt is dissolved in 90.0 g H_2O. This solution has a vapor pressure that is 3.2% lower than that of pure water at the same temperature. What is the salt?
A) LiCl B) NaCl C) KCl D) RbCl E) CsCl

11. A solution consisting of 0.276 mol of methylbenzene, $C_6H_5CH_3$, in 242 g of nitrobenzene, $C_6H_5NO_2$, freezes at –2.0 °C. Pure nitrobenzene freezes at 6.0 °C. What is the freezing-point depression constant of nitrobenzene?
A) 7.2 °C/*m* B) 1.8 °C/*m* C) 14.0 °C/*m* D) 5.3 °C/*m* E) 7.0 °C/*m*

12. A cucumber is placed in a concentrated salt solution. What will most likely happen?
A) Water will flow from the cucumber to the solution.
B) Water will flow from the solution to the cucumber.
C) Salt will flow into the cucumber.
D) Salt will precipitate out.
E) No change will occur.

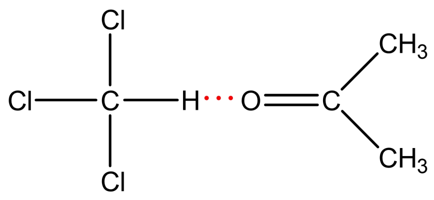
13. The phrase "like dissolves like" refers to the fact that __________.
A) gases can only dissolve other gases
B) polar solvents dissolve nonpolar solutes and vice versa
C) solvents can only dissolve solutes of similar molar mass
D) polar solvents dissolve polar solutes and nonpolar solvents dissolve nonpolar solutes
E) condensed phases can only dissolve other condensed phases

14. A 2.05 *m* aqueous solution of some unknown had a boiling point of 102.1 °C. Which one of the following could be the unknown compound? The boiling-point-elevation constant for water is 0.52 °C/*m*.
A) NaCl B) CH_3OH C) $C_6H_{12}O_6$ D) Na_2CO_3 E) $CaBr_2$

15. A solution contains 30 ppm of some heavy metal and the density of the solution is 1.00 g/mL. This means that ________.
A) 100 g of the solution contains 30 g of heavy metal
B) 100 g of the solution contains 30 mg of heavy metal
C) the solution is 30% by mass of heavy metal
D) there are 30 mg of the heavy metal in 1.0 L of this solution
E) the molarity of the solution is 30 *M*

Part B Short questions (15 points)
16. (5 points) Indicate the principal type of solute–solvent interaction in each of the following solutions and rank the solutions from weakest to strongest solute–solvent interaction: **(a)** NaBr in water, **(b)** CCl_4 in benzene (C_6H_6), **(c)** ethanol (C_2H_5OH) in water.

17. (6 points) At 35 °C the vapor pressure of acetone, $(CH_3)_2CO$, is 360 torr, and that of chloroform, $CHCl_3$, is 300 torr. Acetone and chloroform can form very weak hydrogen bonds between one another; the chlorine atoms on the carbon give the carbon a sufficient partial positive charge to enable this behavior:



A solution composed of an equal number of moles of acetone and chloroform has a vapor pressure of 250 torr at 35 °C.

(a) What would be the vapor pressure of the solution if it exhibited ideal behavior?

(b) Use the existence of hydrogen bonds between acetone and chloroform molecules to explain the deviation from ideal behavior.

(c) Would you expect the same vapor pressure behavior for acetone and chloromethane (CH_3Cl)? Explain.

18. (4 points) A lithium salt used in lubricating grease has the formula $LiC_nH_{2n+1}O_2$. The salt is soluble in water to the extent of 0.036 g per 100.0 g of water at 25 °C. The osmotic pressure of this solution is found to be 0.0751 atm. Assuming that molality and molarity in such a dilute solution are the same and that the lithium salt is completely dissociated in the solution, determine an appropriate value of *n* in the formula for the salt. (R = 0.0821 L-atm/mol-K)

Chapter 13 Homework Answer Sheet

Name: **Student ID:** **Instructor:** **Score:**

Part A Multiple choice (15 points)

1-5______________

6-10_____________

11-15____________

Part B Short questions (15 points)